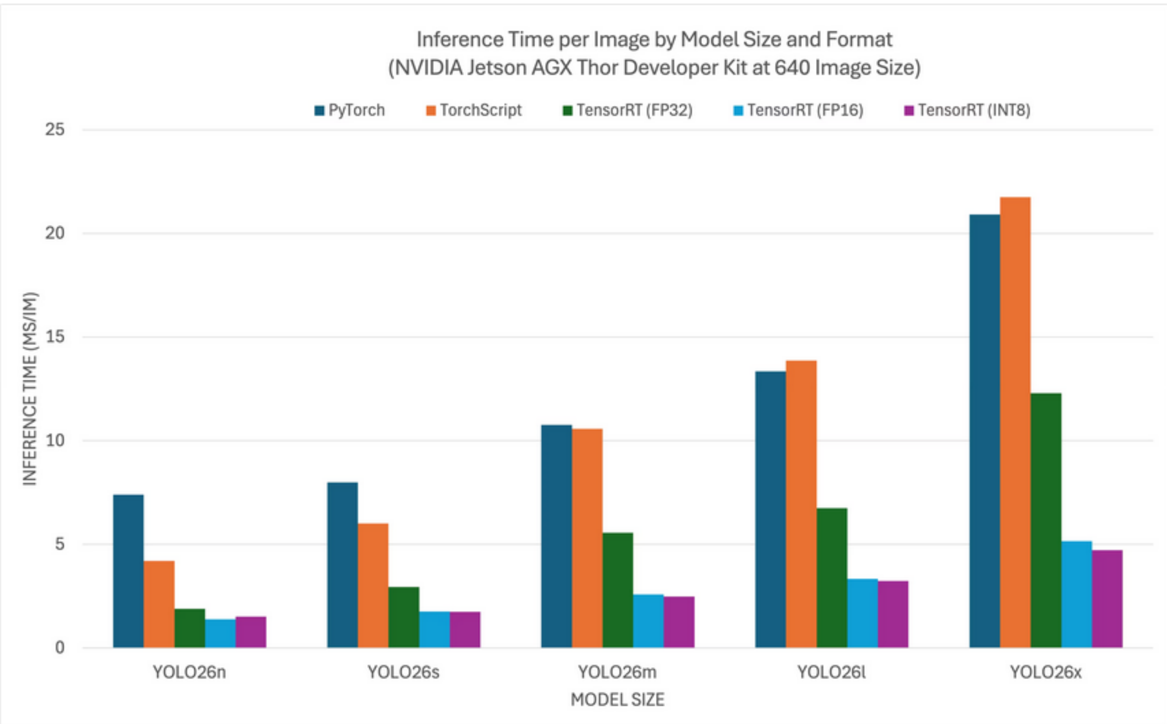
Model Conversion

1、jetson yolo26（Benchmark Test）

yolo26 benchmark test data source from Ultralytics team, testing various model formats (data for reference only).

NVIDIA Jetson AGX Thor Developer Kit



According to the test parameters provided by the Ultralytics team for different format models, we can find that using TensorRT for inference provides the best performance!

Reference link:

```
https://docs.ultralytics.com/guides/nvidia-jetson
```

2、Enable Maximum Board Performance

2.1、Enable MAX Power Mode

Enabling MAX Power Mode on Jetson will ensure all CPU and GPU cores are enabled:

```
sudo nvpmodel -m 0
```

2.2、Enable Jetson Clocks

Enabling Jetson Clocks will ensure all CPU and GPU cores run at maximum frequency:

```
sudo jetson_clocks
```

3、Model Conversion

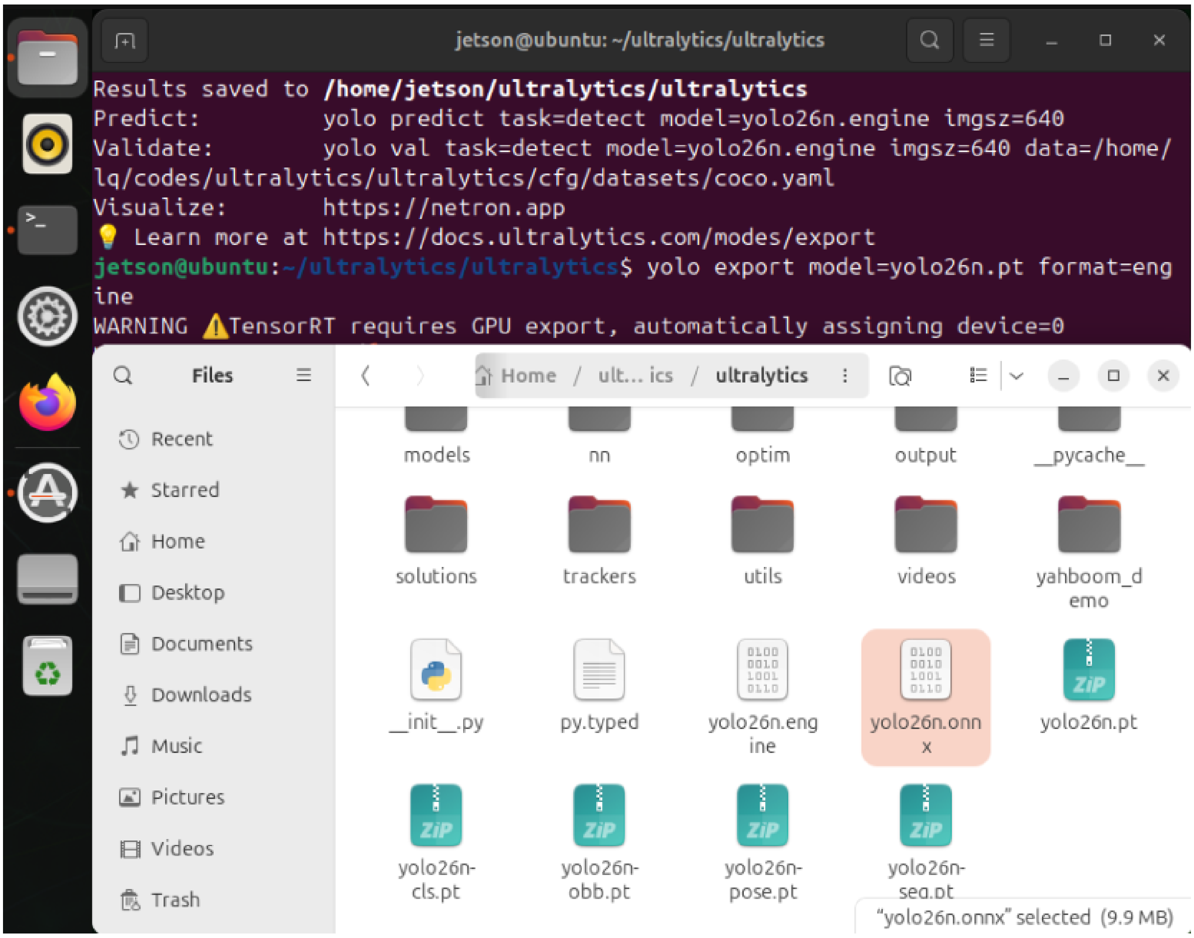
First-time use of yolo26 export mode will automatically install some dependencies, just wait for the automatic completion! Choose one of the two methods: CLI and Python.

3.1、CLI：pt → onnx → engine

Convert PyTorch format model to TensorRT: The conversion process will automatically generate ONNX model

```
cd /home/jetson/ultralytics/ultralytics
```

```
yolo export model=yolo26n.pt format=engine
# yolo export model=yolo26n-seg.pt format=engine
# yolo export model=yolo26n-pose.pt format=engine
# yolo export model=yolo26n-cls.pt format=engine
# yolo export model=yolo26n-obb.pt format=engine
```



3.2、Python：pt → onnx → engine

Convert PyTorch format model to TensorRT: The conversion process will automatically generate ONNX model

```
cd /home/jetson/ultralytics/ultralytics/yahboom_yolo26
```

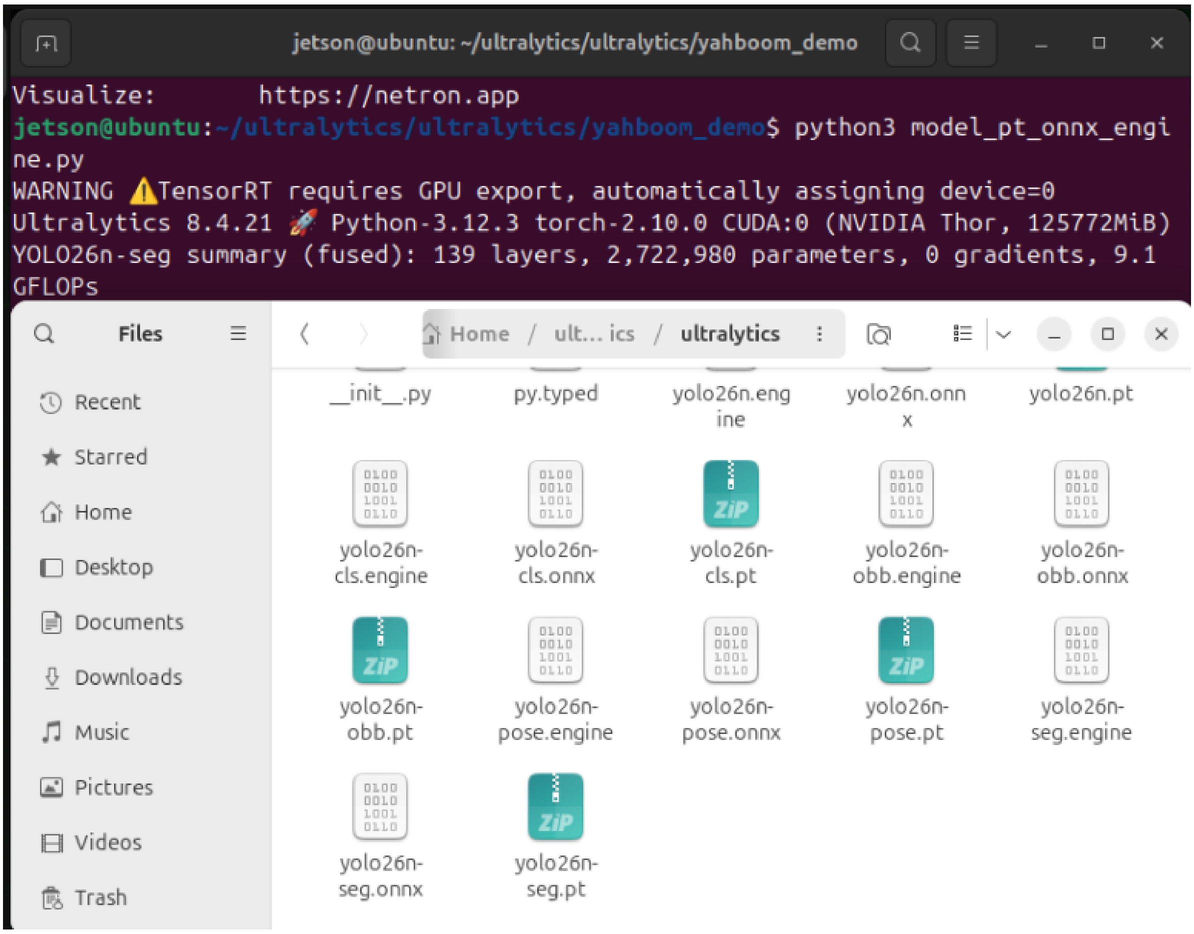
```
python3 model_pt_onnx_engine.py
```

```
from ultralytics import YOLO

# Load a yolo26n PyTorch model
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n.pt")
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-seg.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-pose.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-cls.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n-obb.pt")

# Export the model to TensorRT
model.export(format="engine")
```

Note: The converted model files are located in the converted model file location



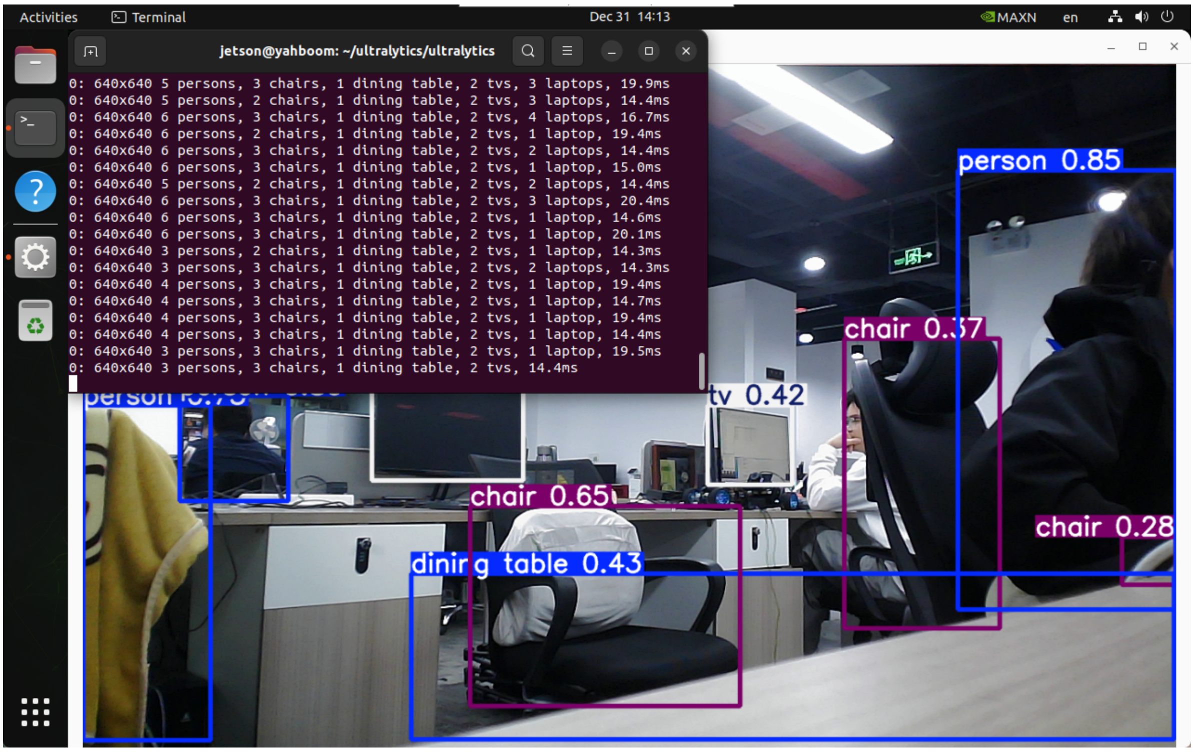
4、Model Prediction

CLI Usage

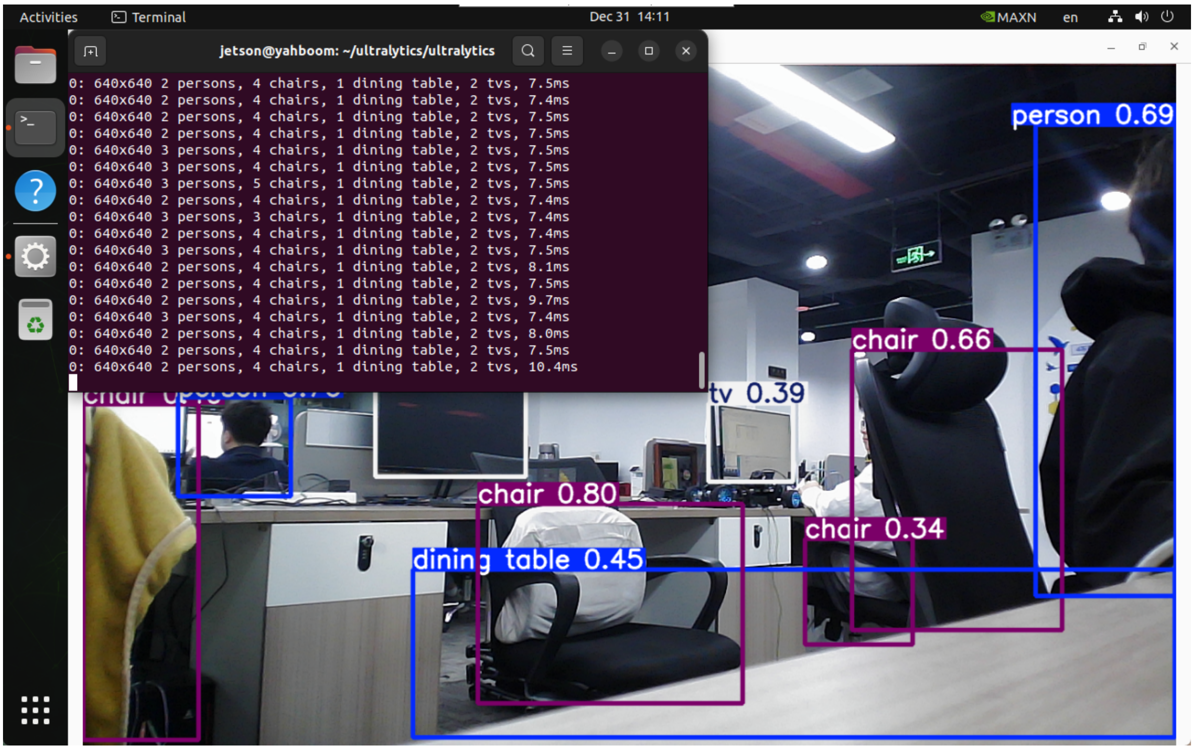
CLI currently only supports calling USB cameras.

```
cd /home/jetson/ultralytics/ultralytics
```

```
yolo predict model=yolo26n.onnx source=0 save=False show
```




```
yolo predict model=yolo26n.engine source=0 save=False show
```



References

https://docs.ultralytics.com/guides/nvidia-jetson/

https://docs.ultralytics.com/integrations/tensorrt/